* The documentation and process conversion *
* measures necessary to comply with this *
* amendment shall be completed by 30 January 1997. *

* INCH POUND *

MIL-S-19500/225E
AMENDMENT 3
30 November 1997
SUPERSEDING
AMENDMENT 2
11 October 1996

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON,
TYPES 2N1711, 2N1711S, 2N1890, 2N1890S, JAN AND JANTX

This amendment forms a part of MIL-S-19500/225E, dated 15 April 1994, and is approved for use by all Department and Agencies of the Department of Defense.

PAGE 3

4.3.2, delete and substitute as follows:

"4.3.2 Thermal impedance $Z_{\Theta JX}$ measurements (for qualification only) The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3131.

a). I_M measurement current.....10 Ma.

b). I_H forward heating current.....50 mA minimum.

c). t_H heating time.....10 ms.

d). t_{MD} measurement delay time.....100 μs minimum.

e). V_{CE} collector - emitter voltage.....10 V dc.

The maximum limit for $Z_{\Theta JX}$ under these test conditions is $Z_{\Theta JX}$ (max) = 58°C/W".

4.3.2.1, delete in its' entirety.

PAGE 4

4.4.1, delete in its entirety and substitute as follows:

"4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The test conditions for $Z_{\Theta JX}$ shall be those used in 4.3.2."

4.4.2, subgroup E2, conditions, delete "V_R" and substitute "V_{CBO}".

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PAGE 5

TABLE I, subgroup 2, thermal impedance test, limits max column, delete "4.5" and substitute "58".

* TABLE I, subgroup 2, collector to emitter saturated voltage, $V_{ce(sat)1}$ and $V_{ce(sat)2}$, limits min column delete “0.2”

PAGE 6

TABLE I, subgroup 2, base emitter saturated voltage, max limits, delete "9.0" and substitute "0.9".

PAGE 7

TABLE I, subgroup 4, small -signal short-circuit output admittance, limits min column, delete "0"; limits max. column, delete "0.5", unit column, delete "µΩ".

PAGE 8

Note 2, delete ".2 ns" and substitute "2 ns".

PAGE 9

4.5.2, a., delete "28 mA" and "47 mA" and substitute "30 mA" and "50 mA" respectively; f., delete "Maximum limit shall be $R_{\Theta JC}$ = 58°C/W." and substitute "t_{md} measurement delay time = 100 µs maximum"; add new note g as follows: "g. I_m measurement current = 10 mA."; add new note h as follows: h. maximum limit shall be $R_{\theta JC}$ = 58°C/W."

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1867)

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